Observable systems with divergent isolated Kalman covariances

A counterexample to the isolated Kalman filtering convergence conjecture

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Abstract

Plain message. The isolated Kalman filter (IsoKF) lets nodes of a network update only the part of a joint covariance that an observation touches, and approximates the rest. Its authors conjectured that, for any completely observable and controllable system, the covariance it reports settles to a finite, unique, stable steady state. This is false. We give a three-node linear system with pairwise observations, observable from a single epoch, for which the reported variances grow without bound, although every local block the filter uses stays positive definite and the ordinary Kalman filter stays bounded.

Results. A sign-cone and harmonic-mean argument proves exponential divergence for $A = 3I$ and, through a Perron-root comparison, for every connected non-bipartite graph of pair-sum observations whenever $a^2 \geq \rho$ of a schedule matrix. Extra attenuation of cross-covariances, including deleting them, does not help. Divergence needs unstable dynamics: for Schur-stable A the reported marginals are always bounded. But very mild instability is enough. With computer-assisted proofs (outward-rounded interval arithmetic) we show that for a^2 in $(1, 1.38]$ the triangle has an invariant region in which variances grow like a^{2k}, reached from the prior $10^{12}I$ at eight certified values of a^2 down to $1 + 10^{-8}$. At $a^2 = 21/20$ the prior I provably converges to a bounded fixed point while $10^{12}I$ diverges, so there is no globally attracting steady covariance.

What the failure means. In high-precision numerical tests of the divergent cases the *actual* estimation error of the IsoKF estimate stays bounded, even while the reported covariance diverges. The failure we exhibit is therefore gross pessimism: the reported covariance loses its meaning as an error measure. It is not divergence of the estimate itself in these tests. We also classify exactly which overlapping participant groups keep every local block positive definite, and we give a bounded regime for anchored relative measurements.

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1 Introduction

In plain English. When several robots or sensors estimate a shared state, a central Kalman filter would keep one big covariance matrix describing how uncertain every quantity is and how errors are correlated. That is expensive to communicate. The isolated Kalman filter of Jung, Luft and Weiss [6] saves communication: when an observation involves a few nodes (the *participants*), only those nodes do a proper update, and every correlation with an outside node is rescaled by a cheap correction factor. The authors conjectured that this approximation always settles down when the system is observable and controllable. We show by explicit example that the reported variances can instead grow without limit, and that, for very mildly unstable dynamics, whether they settle can depend on the starting prior. In our tests the estimate itself does not blow up. What fails is the uncertainty the filter reports, which becomes enormously too pessimistic and therefore useless for gating, fusion or consistency checks.

The conjecture. The journal article states, in its concluding section,

> "The IsoKF paradigm leads to a finite, unique, and stable steady-state covariance Σ_{ikf}^{∞} if the system is completely observable and controllable. The global covariance may not be positive semidefinite." [6, Conjecture 1]

The earlier dissertation formulation [5, printed p. 162] adds that the stacked block matrices relating to output-coupled instances are positive semidefinite. Our main counterexample satisfies

that additional clause: every participant block stays positive definite. The journal admits unstable dynamics explicitly: its discussion of the discrete algebraic Riccati equation notes that a stable steady-state solution can exist "even if the dynamic of the system is unstable" [6]. Our examples are therefore inside the conjecture's stated scope. (A previous draft cited a "Section 5.1.2" locator for this remark; we could not confirm that sub-numbering and have removed it.)

Results at a glance. Table 1 lists every result, how it is established, and what is only numerical.

Table 1: Results at a glance. "Proof" means a written analytic proof; "computer-assisted" means a written argument whose finite hypotheses are checked by a re-runnable interval-arithmetic certificate; "numerical" means evidence only.

Result	Statement in brief	Status
Prop. 3.1	Schur-stable A: reported marginals are bounded; divergence needs instability	Proof
Thm. 4.1	$A = 3I$ triangle of pair sums: $\min_i P_{ii} \geq \frac{5}{4}(\frac{9}{5})^k - \frac{1}{4}$; central KF $\leq 3/4$	Proof (+ exact check of 4 epochs)
Thm. 5.1, Cor. 5.3	Any connected non-bipartite pair-sum graph diverges if $a^2 \geq \rho(M)$; extra attenuation or deletion does not help	Proof
Thm. 6.2	Triangle, $Q = R = I$, $a^2 \in (1, 1.38]$: invariant escape region, growth exactly at rate a^2, $P_k/P_{11} \to 2I - \mathbf{1}\mathbf{1}^\mathsf{T}$	Computer-assisted
Cor. 6.3	Prior $10^{12}I$ enters that region for 8 listed a^2 from $1 + 10^{-8}$ to 1.38	Computer-assisted
Thm. 6.4, Cor. 6.5	$a^2 = 21/20$: $P_0 = I$ converges to a bounded fixed point, $P_0 = 10^{12}I$ diverges; no globally attracting steady covariance	Computer-assisted
Finding 7.1	Actual error covariance of the IsoKF estimate stays bounded while the reported one diverges	Numerical (80 digits)
Finding 7.2	Divergence threshold from $P_0 = I$ is $a^2 \approx 1.1225$; deletion variant close to $2 + \sqrt{5}$ (finite-horizon bias unresolved)	Numerical
Thm. 8.1	Local blocks always stay positive definite iff distinct participant sets share at most one node	Proof (+ exact witness)
Prop. 9.1	Anchored full-coordinate relative measurements give bounded marginals	Proof
App. A, B	Exact singularity witnesses outside the algorithm's domain; a positivity-preserving attenuation that still diverges	Proof (+ exact check)

Reading guide. Section 2 fixes the recurrence and compares it with the published equations. Section 3 explains why only unstable dynamics can diverge. Sections 4–5 give the short complete refutation and its graph extension; a reader who wants only the counterexample can stop there. Section 6 shows that mild instability suffices and that there is no globally attracting steady covariance. Section 7 reports, as numerical findings, what the divergence means for the actual estimation error. Sections 8–9 separate local well-posedness from boundedness and identify a safe regime relevant to robotics. Section 10 places the work in the decentralised-fusion literature, and Section 11 records verification, software scope and open questions. Two appendices keep further exact results that lie off the main line.

Contribution and scope. Generic inconsistency of filters that neglect or approximate correlations, and communication-dependent estimation limits, are known (Section 10). Our contribution is specific to the IsoKF recurrence: the analytic divergence theorem and its graph extension, the certified mild-instability and coexistence results, the exact participant-family classification, and the observation that in our tests IsoKF's marginal-ratio rule diverges at much milder instability than simply deleting cross-covariances. No earlier statement of these results was found in a bounded search (Section 11); this is not proof of priority.

2 The covariance recurrence

This section fixes exactly which mathematical recurrence is analysed and shows that it is the one published in [6].

Partition the state into finitely many nodes, with node dimensions allowed to differ. Write $X \succ 0$ for positive definite, $X \succeq 0$ for positive semidefinite and $X \preceq Y$ when $Y - X \succeq 0$. Let P be a symmetric array of covariance blocks. A prediction is

$$P^- = APA^\mathsf{T} + Q, \tag{2.1}$$

with A and Q block diagonal. For an observation with participant set U, write $B = P[U, U]$, let H_U be its sensitivity matrix and $R \succ 0$ its noise covariance. Whenever $B \succ 0$, set

$$S = H_U B H_U^\mathsf{T} + R, \qquad C = B - BH_U^\mathsf{T} S^{-1} H_U B. \tag{2.2}$$

The participant block is replaced by C. For $i \in U$ and $j \notin U$ the external cross block becomes

$$P_{ij}^+ = D_i P_{ij}, \qquad D_i = C_{ii} B_{ii}^{-1}, \tag{2.3}$$

with the transposed block updated accordingly; blocks between two nonparticipants are unchanged. Equations (2.1)–(2.3), without projection, rejection or numerical regularisation, define the recurrence analysed here.

Fidelity to the published update. Table 2 sets the journal's equations beside ours. The journal's correction factor is the ratio of the participant's posterior to prior *marginal* covariance; it is not a square-root or Cholesky factor, and it is applied with the marginal $B_{ii} = \Sigma_{uu}^{(-)}$, not the joint participant block. Nonparticipants receive zero gain, "$K_o = 0$" [6]. For scalar nodes, as in all our counterexamples, there is no ambiguity of factor order either.

Table 2: The published isolated update [6, eqs. (47)–(50)] and the recurrence used here. $\mathcal{P}$: participants; $\bar{\mathcal{P}}$: nonparticipants.

	Jung, Luft and Weiss (2025)	This paper
cross term	$\breve{\Sigma}_{uv}^{(+)} = \Sigma_{uu}^{(+)}(\Sigma_{uu}^{(-)})^{-1}\Sigma_{uv}^{(-)}$, $u \in \mathcal{P}$, $v \in \bar{\mathcal{P}}$ (48)	$P_{ij}^+ = C_{ii}B_{ii}^{-1}P_{ij}$ (2.3)
factor	$\Lambda_{uu} = \Sigma_{uu}^{(+)}(\Sigma_{uu}^{(-)})^{-1}$ (49)	$D_i = C_{ii}B_{ii}^{-1}$
buffers	$B_{S_u} \leftarrow \Lambda_{uu} B_{S_u}$ (50)	dense array (see below)
participants	joint Kalman update of the stacked participant block	C in (2.2)

Buffered implementation equals the dense array. The published algorithm stores each cross-covariance Σ_{uv} in factored form, as a product of per-node correction buffers that are left-multiplied by Λ_{uu} at an update and by the node's transition matrix at a prediction, and assembles Σ_{uv} only when needed. Each of these operations is a left (or, for the partner node,

right) multiplication of the represented block, so the assembled block obeys exactly the same recursion as the dense entry: $\Sigma_{uv} \mapsto A_u \Sigma_{uv} A_v^\mathsf{T}$ at prediction and $\Sigma_{uv} \mapsto \Lambda_{uu} \Sigma_{uv}$ at an update of u alone. For linear time-invariant models, and in particular $A = aI$, the buffered and dense forms therefore produce identical arrays; the factorisation is a storage device, not a different approximation.

Three properties. *Local well-posedness*: each scheduled B and S is positive definite, so the update can proceed. *Covariance boundedness*: the reported marginals stay bounded. *Statistical consistency*: the reported covariance relates correctly to the actual estimation error. None of these should be substituted for another.

3 Stable dynamics cannot diverge

This short section shows that any divergence counterexample must use unstable (or marginally stable) dynamics, and states what remains open for stable dynamics.

Proposition 3.1. *Assume local well-posedness at every step, block-diagonal $A = \mathrm{diag}(A_i)$ and $Q = \mathrm{diag}(Q_{ii})$. Then for every node i and epoch k,*

$$P_{ii}(k) \preceq A_i^k P_{ii}(0)(A_i^k)^\mathsf{T} + \sum_{t=0}^{k-1} A_i^t Q_{ii} (A_i^t)^\mathsf{T}.$$

In particular, if every A_i is Schur stable, all reported marginals are uniformly bounded, for every observation schedule and every prior.

Proof. A local update gives $C = B - B H_U^\mathsf{T} S^{-1} H_U B \preceq B$, hence $C_{ii} \preceq B_{ii}$ for each participant; nonparticipant marginals are unchanged. Prediction maps P_{ii} to $A_i P_{ii} A_i^\mathsf{T} + Q_{ii}$, a monotone map in the Loewner order. Induction gives the bound, whose right side converges when $\rho(A_i) < 1$. □

Thus the boundedness part of the conjecture holds trivially for stable dynamics, and the question of a *unique* stable steady state for Schur-stable A is untouched by this paper. At $a = 1$ the bound grows only linearly; Section 6 records what is known there.

4 A completely observable pairwise counterexample

This section gives the complete short refutation: one explicit system, two lemmas and a four-line induction.

Theorem 4.1 (Divergence with three scalar nodes)**.** *Consider $x_{k+1} = 3x_k + w_k$ on $\mathbb{R}^3$ with $\mathrm{Cov}(w_k) = I_3$ and independent initial covariance $P_0 = I_3$. At each epoch predict once, then process the scalar observation channels*

$$H = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad R = I_3, \tag{4.1}$$

in the displayed order, each on its two nonzero nodes. Let P_k be the IsoKF array after the three updates at epoch k and $m_k = \min_i (P_k)_{ii}$. Every pairwise participant block remains positive definite, every innovation is positive, and

$$m_k \;\geq\; \frac{5}{4}\Big(\frac{9}{5}\Big)^k - \frac{1}{4} \;\longrightarrow\; \infty. \tag{4.2}$$

The system is completely observable, and $(A, Q^{1/2}) = (3I, I)$ is controllable. The centralised Kalman posterior satisfies $P_k^{\mathrm{KF}} \preceq (H^\top H)^{-1} = I_3 - \frac{1}{4}\mathbf{1}\mathbf{1}^\top$ for $k \geq 1$, so every centralised marginal is at most $3/4$.

(Earlier drafts added a deterministic input $u_k = 0$ with input matrix I to make input controllability explicit. Controllability from the process noise suffices for the conjecture, so the device is dropped.)

Lemma 4.2 (An invariant cone)**.** *For scalar nodes write $p_i = P_{ii}$ and $P_{ij} = -c_{ij}$ for $i \neq j$, and consider arrays with*

$$p_i > 0, \qquad 0 \leq c_{ij} < \min(p_i, p_j). \tag{4.3}$$

A prediction with $A = aI$, $a > 0$, and strictly positive diagonal Q preserves (4.3). So does every pair-sum observation $x_i + x_j$ with noise variance $r > 0$.

Proof. Prediction maps $c_{ij} \mapsto a^2 c_{ij}$ and $p_i \mapsto a^2 p_i + Q_{ii}$; strictness is kept because $a^2 c_{ij} < a^2 p_i < a^2 p_i + Q_{ii}$. For a pair update put $p = p_i$, $q = p_j$, $c = c_{ij}$, $u = p - c > 0$, $v = q - c > 0$. Then

$$s = u + v + r > 0, \qquad p' = p - \frac{u^2}{s}, \quad q' = q - \frac{v^2}{s}, \quad c' = c + \frac{uv}{s}, \tag{4.4}$$

so $p' - c' = ur/s > 0$, $q' - c' = vr/s > 0$, $0 < p' \leq p$ and $0 < q' \leq q$. For an outside node ℓ, $c'_{i\ell} = (p'/p)c_{i\ell} < p'$ and $c'_{i\ell} \leq c_{i\ell} < p_\ell$; likewise for j. Every two-node block is positive definite because $c_{ij}^2 < p_i p_j$. □

Lemma 4.3 (Harmonic lower bound)**.** *Under Lemma 4.2, a pair-sum update obeys $p', q' \geq pq/(p+q)$.*

Proof. From (4.4),

$$p' - \frac{pq}{p+q} = \frac{c\{2pq - c(p+q)\} + p^2 r}{(p+q)(p+q-2c+r)} \geq 0, \tag{4.5}$$

since $c \leq \min(p, q) \leq 2pq/(p+q)$. Exchange p and q for the other bound. □

Proof of Theorem 4.1. $P_0 = I$ satisfies (4.3), and Lemma 4.2 gives local well-posedness. If all diagonal entries before the observations are at least t, monotonicity of $pq/(p+q)$ gives the successive lower bounds $(t,t,t) \to (t/2, t/2, t) \to (t/2, t/3, t/3) \to (t/5, t/3, t/5)$ after rows 12, 23, 31. Prediction gives $t = 9m_k + 1$, so $m_{k+1} \geq (9m_k + 1)/5$, and with $m_0 = 1$ induction gives (4.2). $\det H = 2$ gives observability from one epoch. For the centralised filter, $P_k^{\mathrm{KF}} = \{(P_k^{\mathrm{KF},-})^{-1} + H^\top H\}^{-1} \preceq (H^\top H)^{-1}$ and $H^\top H = I + \mathbf{1}\mathbf{1}^\top$. □

Unbounded marginals preclude a finite steady-state covariance, so the example refutes the universal claim of [6, Conjecture 1], with only pairwise observation blocks, all of them positive definite. The bound $9/5$ is loose: exact arithmetic gives $m_k = 2.38, 14.97, 114.2, 982.0$ for $k = 1, \ldots, 4$, a ratio near 8.6 that approaches the open-loop rate $a^2 = 9$; Theorem 6.2(c) proves rate exactly a^2 in a milder range.

5 A schedule-dependent extension

This section turns the mechanism into a criterion for any connected graph of pair-sum observations and shows that additional cross-covariance attenuation cannot repair it.

For an observed pair ij let T_{ij} replace entries i, j of a vector by their sum and leave the others unchanged. For a repeated ordered schedule of L pairs set $M = T_{i_L j_L} \cdots T_{i_1 j_1}$.

Theorem 5.1 (Graph criterion)**.** *Let the union observation graph be connected, $A = aI$ with $a > 0$, Q positive diagonal with $Q \succeq q_* I$, $q_* > 0$, the prior positive diagonal, and the observations unit pair sums $x_i + x_j$ with positive noise variances. Let $\rho = \rho(M)$ with Perron vector $v > 0$, $\min_i v_i = 1$. Then $t_k := \min_i v_i (P_k)_{ii}$ satisfies*

$$t_{k+1} \geq \frac{a^2 t_k + q_*}{\rho}. \tag{5.1}$$

Hence all marginals diverge if $a^2 \geq \rho$: at least exponentially if $a^2 > \rho$, at least linearly if $a^2 = \rho$. If the graph is also non-bipartite, the stacked observation matrix has full column rank, giving a completely observable and controllable counterexample on every such graph.

Proof. For $z_i = 1/p_i$, Lemma 4.3 gives $z^+ \leq T_{ij} z$ componentwise. The T_{ij} are nonnegative, so $z^{\text{out}} \leq M z^{\text{in}}$ over a schedule; each $T_{ij} \geq I$ and contains its edge, so connectedness makes M irreducible with a positive Perron vector. Before the observations $p_i^- \geq (a^2 t_k + q_*)/v_i$, i.e. $z_i^- \leq v_i/(a^2 t_k + q_*)$; multiplying by M and inverting gives (5.1). If $Hx = 0$ then $x_i = -x_j$ on every edge; an odd cycle forces $x = 0$. □

For the triangle schedule of Theorem 4.1,

$$M = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{pmatrix}, \qquad \rho = 2 + \sqrt{5}, \qquad v = (\varphi, 1, \varphi)^{\mathsf{T}}, \quad \varphi = \tfrac{1+\sqrt{5}}{2},$$

so $a^2 \geq 2 + \sqrt{5} \approx 4.236$ suffices.

Remark 5.2 (Scope of Theorem 5.1). (i) *Order.* All six orders of the three triangle edges give the same ρ: cyclic shifts give conjugate products ($\rho(XY) = \rho(YX)$), and any order is mapped to any other by relabelling the nodes. (ii) *Robustness.* $a^2 > \rho$ is an open condition, so divergence persists under small changes of a, and the theorem allows every positive choice of noise variances and diagonal process noise; it does not claim robustness to arbitrary structural perturbations of the measurements. (iii) *Coefficients.* The theorem is stated for unit pair sums. A row $\kappa_e(d_i x_i + d_j x_j)$ with $\kappa_e > 0$ and positive node weights d_i reduces to it: in the coordinates $y_i = d_i x_i$, $A = aI$ is preserved, Q stays positive diagonal, R_e becomes R_e/κ_e^2, and the recurrence is equivariant under diagonal scaling because the scalar ratio D_i is scale invariant.

Corollary 5.3 (Attenuation and deletion cannot restore boundedness)**.** *Keep the local update (2.2) but allow any adaptive factors $0 \leq \eta_{i\ell} \leq 1$ in $P_{i\ell}^+ = \eta_{i\ell}(p_i'/p_i) P_{i\ell}$ for participants i and outside nodes ℓ. The conclusions of Theorems 4.1 and 5.1 still hold. This includes $\eta \equiv 0$, i.e. deleting all external cross-covariances, and the common-factor repair of Appendix B.*

Proof. The participant posterior p_i' depends only on the participant block, so it is unchanged by η. The new outside magnitude lies in $[0, (p_i'/p_i) c_{i\ell}]$, so the strict cone (4.3) is preserved as in Lemma 4.2. Lemma 4.3 holds for every c in the cone, so every lower bound applies verbatim. □

Corollary 5.3 is an impossibility result for this class of corrections, not for decentralised filtering in general. The bound is only sufficient: numerically, deletion is bounded up to a threshold close to $2 + \sqrt{5}$, while the IsoKF rule itself diverges much earlier (Finding 7.2).

Why local validity does not prevent the failure. The array $E = 2I_3 - \mathbf{1}\mathbf{1}^{\mathsf{T}}$ has positive diagonal and every two-node block equals $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right) \succeq 0$, which annihilates its pair-sum row. Yet H is invertible, and E has eigenvalues $2, 2, -1$. Locally compatible but globally inconsistent covariance relations let the approximation ignore a globally informative set of measurements. Theorem 6.2(d) proves that the normalised array indeed converges to E in the mild-instability range.

6 Mild instability suffices, and no steady covariance attracts globally

This section shows, with computer-assisted proofs, that divergence does not need strong instability, and that whether the covariance settles can depend on the prior. Throughout, $A = aI$, $Q = R = I$ and the triangle schedule of Theorem 4.1 are used. Scaling $P \mapsto P/q$ maps $(Q, R) = (qI, qI)$ to (I, I), so all statements hold for $Q = R = qI$ with every covariance multiplied by q; general $Q \neq R$ is not covered.

6.1 Deficit form of the recursion

On the cone (4.3) define six ordered *deficits* $e_{ij} = p_i - c_{ij} > 0$, $i \neq j$, and let ℓ be the third node of $\{i, j\}$.

Lemma 6.1. *Prediction maps $p \mapsto a^2 p + 1$ and $e \mapsto a^2 e + 1$. The pair-sum update on $\{i, j\}$ with $r = 1$, writing $u = e_{ij}$, $v = e_{ji}$, $s = u + v + 1$, $\lambda_i = u^2/s$, $\lambda_j = v^2/s$, is*

$$p_i' = p_i - \lambda_i, \quad e_{ij}' = \frac{u}{s}, \quad e_{i\ell}' = \Big(1 - \frac{\lambda_i}{p_i}\Big) e_{i\ell}, \quad e_{\ell i}' = e_{\ell i} + \Big(1 - \frac{e_{i\ell}}{p_i}\Big)\lambda_i, \tag{6.1}$$

and symmetrically for j; all other quantities are unchanged. Moreover $e_{12} - e_{21} + e_{23} - e_{32} + e_{31} - e_{13} = 0$ for every array.

Proof. The first two formulas are (4.4) with $p' - c' = u/s$. By (2.3), $c_{i\ell}' = (p_i'/p_i) c_{i\ell}$, so $e_{i\ell}' = (p_i'/p_i) e_{i\ell}$ and $e_{\ell i}' = p_\ell - c_{i\ell}' = e_{\ell i} + (c_{i\ell}/p_i)\lambda_i$ with $c_{i\ell}/p_i = 1 - e_{i\ell}/p_i$. The cycle identity holds since $e_{ij} - e_{ji} = p_i - p_j$. □

The update depends on p only through e/p and λ/p, so for large p the deficits evolve almost autonomously. Our certificates use (6.1) with e_{13} eliminated by the cycle identity; computing deficits as differences of large numbers, or iterating without the identity, produces spurious numerical modes.

6.2 Large priors diverge under mild instability

Certificate C1. The range $a^2 \in [1, 1.38]$ is split into 190 intervals I_m of width $1/500$. For each m a box $B_m \subset \mathbb{R}^5$ of radius 0.006 in the deficit coordinates $(e_{12}, e_{21}, e_{23}, e_{32}, e_{31})$ is given. With $\theta_i = 1/p_i$, the certificate verifies in outward-rounded interval arithmetic, for every $a^2 \in I_m$ and every value of each $\theta_i \in [0, \bar{\theta}]$, where $\bar{\theta} \geq 10^{-6}$ is the upper endpoint of an outward-rounded enclosure of 10^{-6} (each occurrence independently): (1) $F^4(B_m) \subset \operatorname{int} B_m$, where F is one epoch of (6.1), using a mean-value form whose interval Jacobian is propagated by the chain rule and intersected with naive interval evaluation; (2) every innovation within each epoch is positive, and all enclosed deficits, including e_{13}, are positive at each epoch end (positivity within an epoch follows from Lemma 4.2); (3) in each of the four epochs each node's total loss Λ_i over its two updates obeys $L_m^- \leq \Lambda_i \leq L_m$, with $\max_m L_m \leq 3.1717$ and $L_1^- \geq 2.0104$ on $I_1 = [1, 1.002]$. The mean-value form is valid because, with the realised θ-values along a true trajectory held fixed, the epoch map is C^1 in (x, a^2) on the convex set $B_m \times I_m$.

Theorem 6.2 (Mild-instability divergence; computer-assisted)**.** *Let $Q = R = I$, $a^2 \in (1, 1.38]$, $a^2 \in I_m$, and $\pi_a = \max\{10^6, (L_m - 1)/(a^2 - 1)\}$. Suppose that at some epoch end k_0 the IsoKF array lies in the cone (4.3), has deficits in B_m and has $\min_i p_i \geq \pi_a$. Then for all $k \geq k_0$:*

(a) every local block and innovation stays positive definite, and the deficits at epochs $k_0 + 4n$ lie in B_m;

(b) $p_i(k) - \pi_a \geq a^{2(k-k_0)}(p_i(k_0) - \pi_a)$ *and* $p_i(k) \leq a^{2(k-k_0)}(p_i(k_0) + \frac{1}{a^2-1})$*;*

(c) if $\min_i p_i(k_0) > \pi_a$*, then* $\kappa = \lim_k a^{-2k} p_i(k)$ *exists, is positive and is the same for all three nodes, and* $p_i(k) = \kappa a^{2k}(1 + O(a^{-2k}))$*;*

(d) under the strict inequality in (c), $a^{-2k}P_k \to \kappa E$ *with* $E = 2I_3 - \mathbf{1}\mathbf{1}^\mathsf{T}$*; equivalently* $P_k/(P_k)_{11} \to E$*, an indefinite array whose principal block on each observed pair annihilates that pair's local observation row.*

Proof. (a),(b) by induction over blocks of four epochs. At a block start with deficits in B_m and $\min p \geq \pi_a$: after prediction $p_i \geq a^2\pi_a + 1$, and after any part of the epoch's losses $p_i \geq a^2\pi_a + 1 - L_m \geq \pi_a \geq 10^6$ since $(a^2-1)\pi_a \geq L_m - 1$. So every realised $\theta_i \in [0, 10^{-6}]$ and the state stays in the certified enclosures, where losses are at most L_m and denominators are positive. Hence $p_i(k+1) \geq a^2 p_i(k) + 1 - L_m$, i.e. $p_i(k+1) - \pi_a \geq a^2(p_i(k) - \pi_a)$. By C1(1) the next block starts in B_m; positivity is Lemma 4.2. The upper bound uses $\Lambda_i \geq 0$. (c) $a^{-2(k+1)}p_i(k+1) = a^{-2k}p_i(k) + a^{-2(k+1)}(1 - \Lambda_i(k))$ with $|1 - \Lambda_i| \leq \max(1, L_m)$, so the limit exists with error $O(a^{-2k})$ and, by (b), is at least $a^{-2k_0}(p_i(k_0) - \pi_a) > 0$. The limits coincide because $|p_i - p_j| = |e_{ij} - e_{ji}|$ is bounded on the certified hull. (d) $c_{ij} = p_i - e_{ij}$ with e bounded. □

Corollary 6.3 (Entry from a positive definite prior; computer-assisted)**.** *Certificate C2 is an outward-rounded interval simulation (60 digits) of the recursion from* $P_0 = 10^{12}I$*. It verifies the hypotheses of Theorem 6.2, with* $\min p > \pi_a$ *strictly (lower interval endpoints compared exactly with the rational* π_a*), at epoch* k_0 *for*

$$a^2 \in \{1 + 10^{-8},\ 1 + 10^{-6},\ 1.001,\ 1.01,\ 21/20,\ 1.1,\ 1.2,\ 1.38\},$$
$$k_0 \in \{29, 29, 29, 29, 31, 32, 35, 43\} \text{ respectively.}$$

For each of these values, $P_0 = 10^{12}I$ *(and* $10^{12}qI$ *when* $Q = R = qI$*) gives* $(P_k)_{ii} \sim \kappa a^{2k} \to \infty$ *and* $P_k/(P_k)_{11} \to E$*. The same holds on some open neighbourhood of* $10^{12}I$*, because* $P_0 \mapsto P_{k_0}$ *is continuous (finitely many steps with positive innovations and marginals) and the hypotheses at* k_0 *are strict inequalities.*

Exactly what is and is not proved here. The invariant escape region is certified for *every* $a^2 \in (1, 1.38]$, but entry from an explicit prior is certified only at the eight listed values; a uniform entry argument through the $O(\log c)$-epoch transient was not obtained, and the neighbourhood in Corollary 6.3 is not explicit. Nothing is proved here for $a^2 \in (1.38, 2 + \sqrt{5})$, which is covered by neither this result nor Theorem 5.1. At $a = 1$, a trajectory starting in B_1 cannot remain indefinitely in the certified regime $\min_i p_i \geq 10^6$: while it does, C1 gives $\Lambda_i \geq 2.0104$, so each marginal decreases by at least 1.01 per epoch. Global boundedness at $a = 1$ is not established here.

6.3 Coexistence: a bounded and a divergent trajectory

Theorem 6.4 (Bounded trajectory at $a^2 = 21/20$; computer-assisted)**.** *Let* $a^2 = 21/20$*,* $Q = R = I$*,* $P_0 = I$*. Then* P_k *converges to a fixed point* P_* *of the epoch map. The fixed point lies in the certificate box* B_{cert} *(centre recorded to 25 digits, radius* 10^{-6} *in each coordinate; exact endpoints in the certificate output), and hence in the outward-rounded box*

$$p_1 \in [1.4923145, 1.4923166], \quad p_2 \in [1.6897335, 1.6897356], \quad p_3 \in [1.4282154, 1.4282175],$$
$$c_{12} \in [1.0011916, 1.0011937], \quad c_{23} \in [1.0904812, 1.0904833], \quad c_{31} \in [1.1027373, 1.1027394].$$

In particular $\sup_k \max_i (P_k)_{ii} < \infty$*. (*$P_*$ *is itself indefinite:* $\mathbf{1}^\mathsf{T} P_* \mathbf{1} < 0$*.)*

Proof. Certificate C3 is an interval simulation (250 digits) of (2.1)–(2.3) in (p, c) variables from $P_0 = I$. Write $B = B_{\text{cert}}$; the larger displayed box is used only to report bounds and is not claimed to be invariant. C3 shows that P_{80} lies in the interior of B, and, by mean-value enclosures, that $F(B) \subset B$, $F^4(B) \subset \operatorname{int} B$ and $\|D(F^4)\|_\infty \le 0.5955 < 1$ on B (row sums of the interval Jacobian's absolute values computed in interval arithmetic, upper endpoints used), with all innovations and marginals positive and B inside the cone. By the mean-value theorem F^4 is an ∞-norm contraction of B into itself, so it has a unique fixed point $x_* \in B$ attracting all of B. Since $F(x_*) \in B$ is also fixed by F^4, $F(x_*) = x_*$. Finally $P_{80+4n+j} = F^j(F^{4n}(P_{80})) \to F^j(x_*) = x_*$ for $j = 0, \dots, 3$. □

Corollary 6.5 (No unique globally attracting steady covariance)**.** *At $a^2 = 21/20$ with $Q = R = I$, the prior $P_0 = I$ gives a bounded convergent covariance (Theorem 6.4), while $P_0 = 10^{12} I$, and every prior in an open neighbourhood of it, gives marginals growing like $\kappa(21/20)^k$ (Theorem 6.2, Corollary 6.3). The centralised Riccati recursion for the same system converges from every $P_0 \succeq 0$ to the unique stabilising solution, because (A, H) is observable and $(A, Q^{1/2})$ is controllable [14].*

So the conjecture's "unique, stable" steady covariance fails to attract globally even where small priors do not diverge, at an instability of only 5% in variance per epoch. For $a^2 \in \{1 + 10^{-8}, 1 + 10^{-6}, 1.001, 1.01\}$ the divergent prior is certified; the bounded trajectory from $P_0 = I$ at those values is numerically convergent but not separately certified. Figure 1 shows the two behaviours.

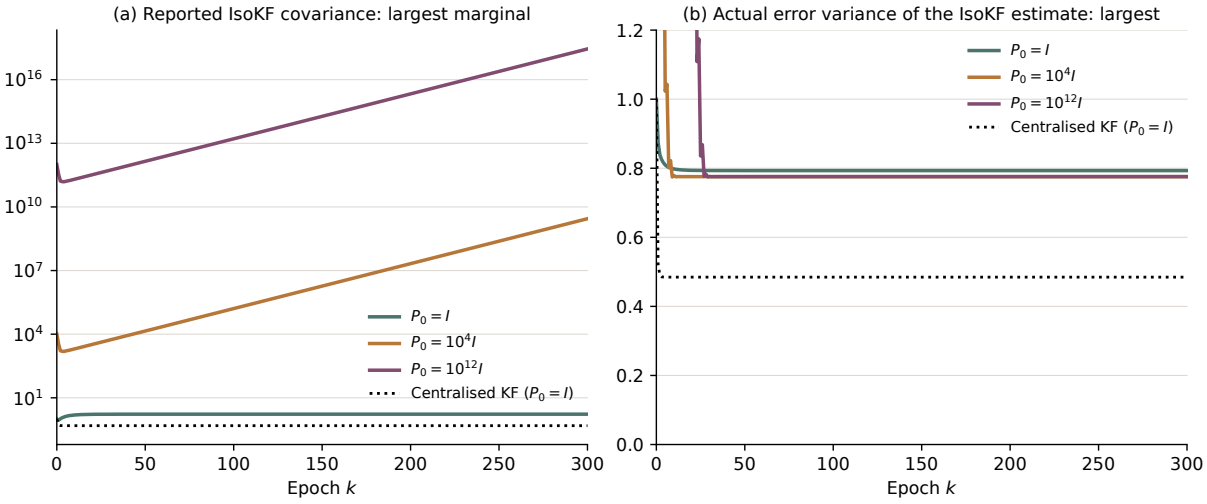


Figure 1: Triangle schedule, $a^2 = 21/20$, $Q = R = I$, epochs 0–300, computed with 150-digit decimal arithmetic. (a) Largest reported IsoKF marginal for three priors (logarithmic scale): $P_0 = I$ settles (proved, Theorem 6.4); $P_0 = 10^{12} I$ grows like 1.05^k (proved, Corollary 6.3); $P_0 = 10^4 I$ also grows (numerical only). Dotted: centralised Kalman filter. (b) Largest variance of the *actual* error of the IsoKF estimate for the same priors (linear scale; numerical, Finding 7.1): it stays below 0.8 in all three cases although the reported variance in (a) reaches 10^{17}.

7 What the divergence means: numerical findings

This section is numerical. It answers two practical questions: does the IsoKF estimate itself go wrong, and how unstable must the system be? None of it is used in any proof.

Estimator model. At a pair observation $y = x_i + x_j + v$ the participants update their estimates with the local gain $K = Bh^\mathsf{T}/S = (u, v)^\mathsf{T}/s$ computed from the reported block; nonparticipants keep their estimates ($K_o = 0$); prediction gives $\hat{x} \leftarrow A\hat{x}$. The gains are deterministic, so

the covariance Σ of the true error obeys the exact linear recursion $\Sigma \leftarrow A\Sigma A^{\mathsf{T}} + Q$, then $\Sigma \leftarrow F\Sigma F^{\mathsf{T}} + KK^{\mathsf{T}}$ with $F = I - Kh$ embedded in three dimensions, from $\Sigma_0 = P_0$. No Monte Carlo is involved; we evaluate it in 80-digit arithmetic.

Numerical finding 7.1 (Bounded actual error, divergent reported covariance). In the tested divergent cases the actual error covariance appears to converge to a bounded limit while the reported covariance diverges (the bounded case $a^2 = 1.05$, $P_0 = I$ is included for comparison). The computation needs high precision: in double precision the $a^2 = 9$ error recursion blows up spuriously through cancellation in $p - c$.

case	reported $\max_i P_{ii}$	actual error variances (limit)	centralised KF
$a^2 = 9$, $P_0 = I$ (Thm. 4.1)	1.0×10^{190} at $k = 200$	(4.25, 3.25, 3.25)	0.678
$a^2 = 1.05$, $P_0 = I$	1.690 (bounded)	(0.556, 0.793, 0.512)	0.485
$a^2 = 1.05$, $P_0 = 10^4 I$	2.9×10^{45} at $k = 2000$	(0.561, 0.776, 0.496)	0.485
$a^2 = 1.01$, $P_0 = 10^8 I$	5.4×10^{15} at $k = 2000$	(0.556, 0.765, 0.493)	0.482
$a^2 = 2$, $P_0 = I$	5.1×10^{149} at $k = 500$	(0.665, 1.041, 0.558)	0.542

The gains stay of order one because they depend only on the bounded deficits, and the asymptotic epoch error-transition matrix has spectral radius below one (≈ 0 for $a^2 = 9$, 0.45–0.57 otherwise).

Interpretation. In these tests the failure is gross pessimism, not estimate divergence: the reported covariance stops meaning anything as a measure of error, while the estimate keeps a bounded error. The estimate is nevertheless worse than the centralised one: by up to about 90% per node in the mild cases and by a factor of about five to six for $a^2 = 9$. A reported covariance that is 10^{45} times too large is useless for outlier gating, for fusion with other sources, and for any consistency check, and it would make a downstream consumer ignore the estimate. We have not proved boundedness of the actual error; the observed limit $(17/4, 13/4, 13/4)$ for $a^2 = 9$ is not derived analytically.

Numerical finding 7.2 (Thresholds). Double-precision runs of 20000 epochs with bisection tolerance 10^{-4} (classification by convergence, by $\min p > 10^{12}$, or by sustained growth; brackets near slow critical dynamics are resolution-limited) give the following a^2-brackets for the onset of divergence on the triangle with $Q = R = I$:

rule	prior	threshold bracket for a^2
IsoKF (2.3)	I	[1.12251, 1.12256]
IsoKF	$3I$	[1.12241, 1.12246]
IsoKF	$10I$	[1.12206, 1.12211]
IsoKF	$100I$	[1.05207, 1.05212]
IsoKF	$10^4 I$	[1.00081, 1.00086]
deletion ($P_{i\ell}^{+} = 0$)	I, $10I$	[4.2341, 4.2342]
deletion	$10^4 I$	[4.2357, 4.2358]

For the IsoKF rule the threshold falls towards 1 as the prior grows, consistent with Theorem 6.2, where the required prior size scales like $1/(a^2 - 1)$. For deletion the estimates are close to the Perron threshold $2 + \sqrt{5} \approx 4.2361$ of Theorem 5.1, with an unresolved finite-horizon bias (the brackets do not contain $2 + \sqrt{5}$, and dynamics near the threshold are very slow). The IsoKF marginal-ratio rule is thus markedly *less* robust than simply discarding cross-covariances: the divergence is not merely the generic effect of neglecting correlations, but a property of this rule.

8 Which participant families are always locally well defined?

This section separates local well-posedness from boundedness: it classifies exactly when every local block stays positive definite, which, by Theorem 4.1, still does not imply convergence.

Let $\mathcal{G}$ be a finite family of distinct nonempty participant sets, node dimensions arbitrary. The admissible cone $\mathcal{A}_{\mathcal{G}}$ consists of symmetric arrays whose node diagonal blocks and principal blocks $P[U,U]$, $U \in \mathcal{G}$, are positive definite; such an array need not be globally positive definite.

Theorem 8.1 (Participant-family criterion)**.** *The following are equivalent.*

1. *For every $P \in \mathcal{A}_{\mathcal{G}}$, every observation on some $U \in \mathcal{G}$ with positive definite noise and every measurement matrix H_U (of any rank, including rows that do not touch every participant), the update (2.2)–(2.3) preserves $\mathcal{A}_{\mathcal{G}}$.*
2. *Any two distinct $U, V \in \mathcal{G}$ satisfy $|U \cap V| \le 1$.*

Under condition 2, block-diagonal predictions with positive definite node noise preserve admissibility, so every finite schedule remains well defined.

For participant pairs, a closely related sufficiency statement is due to Luft et al. [10, Proposition 1]: if every two-robot block is positive definite, their update (eq. (26), and the scaled eq. (28) with $|\lambda| \le 1$) keeps every two-robot block positive definite. Theorem 8.1 adds the converse and participant groups of any size.

Proof. Sufficiency. Fix an update on U; then $0 \prec C \preceq B$. A target group V disjoint from U is unchanged and $V = U$ becomes C. If $U \cap V = \{i\}$, write the old block on V as $\left(\begin{smallmatrix} X & Y \\ Y^{\mathsf{T}} & Z \end{smallmatrix}\right) \succ 0$ with $X = P_{ii}$, and set $X^+ = C_{ii}$, $F = X^+X^{-1}$. The updated block is

$$\begin{pmatrix} F & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} X & Y \\ Y^{\mathsf{T}} & Z \end{pmatrix} \begin{pmatrix} F^{\mathsf{T}} & 0 \\ 0 & I \end{pmatrix} + \begin{pmatrix} X^+ - X^+X^{-1}X^+ & 0 \\ 0 & 0 \end{pmatrix},$$

a positive definite congruence plus a positive semidefinite term ($0 \prec X^+ \preceq X$). Predictions act by congruence plus positive noise.

Necessity. If distinct groups share two nodes, orient them so $V \setminus U \neq \emptyset$ and pick $a, b \in U \cap V$, $c \in V \setminus U$. On these coordinates take

$$P_0 = \begin{pmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -1/2 \\ 1/2 & -1/2 & 1 \end{pmatrix} \succ 0$$

(leading minors $1, 1, 1/2$), other coordinates independent with unit variance. Observe $x_a - x_b$ on U with variance $1/4$:

$$P_1 = \begin{pmatrix} 5/9 & 4/9 & 5/18 \\ 4/9 & 5/9 & -5/18 \\ 5/18 & -5/18 & 1 \end{pmatrix}, \qquad \det P_1 = -7/162,$$

so the block on V is no longer positive definite. If U has further participants or node coordinates, add independent observation rows for them with positive noise; they do not change P_1. This is why condition 1 must allow measurement rows that leave some participants untouched by the critical row. □

Every pairwise family satisfies condition 2, as do singletons and a single large group. The obstruction concerns distinct groups sharing two or more nodes; it does not say every such schedule fails for every prior. With $h = (2, -2, -1)^{\mathsf{T}}$, $P_1 h = (-1/18, 1/18, 1/9)^{\mathsf{T}}$ and $h^{\mathsf{T}} P_1 h = -1/3$, so a further observation on V with noise $1/3$ has zero innovation and a nonzero gain numerator. In hypergraph language condition 2 says $\mathcal{G}$ is a *linear* hypergraph.

9 A bounded regime with direct anchors

This section identifies where the negative results do not bite, and explains which measurement models carry the risk in robotics.

Proposition 9.1 (Anchored relative observations)**.** *Consider node states of equal dimension; each period has one block-diagonal prediction and finitely many private and pairwise updates, with initially positive definite participant blocks and positive definite node process noise, and* $\|A_i\|_2 \le b$, $Q_{ii} \preceq qI$ *uniformly. An anchor receives a full-coordinate direct observation every period with noise at most* $r_0 I$*, and each edge of a fixed connected graph receives a full-coordinate difference observation* $x_i - x_j$ *every period with noise at most* rI*; other private or pairwise observations are allowed. Put* $B_0 = r_0$, $B_{d+1} = b^2 B_d + q + r$*. A node at graph distance* d *from the anchor has* $P_{ii}(k) \preceq B_d I$ *at the end of every period* $k \ge d+1$*. Hence, with* D *the largest distance from the anchor, every marginal is bounded by* $\max_{d \le D}\{B_d, b^2 B_d + q\} I$ *for all* $k \ge D+1$*, and is finite during the initial transient.*

Proof. Theorem 8.1 gives local positive definiteness. For a pair prior $\left(\begin{smallmatrix} X & E \\ E^\mathsf{T} & Y \end{smallmatrix}\right)$ and observation $[I, -I]$ put $S = X + Y - E - E^\mathsf{T} + R \succ 0$. Expanding with $X + R - E = S - (Y - E^\mathsf{T})$ gives the exact identity $X + R - C_{jj} = (X + R - E)S^{-1}(X + R - E^\mathsf{T}) \succeq 0$, so $C_{jj} \preceq X + R$ without commutativity assumptions. The anchor's posterior is at most $r_0 I$ every period, and later updates only decrease participating marginals. If a neighbour at distance d has end-of-period bound $B_d I$, then in the next period it is at most $(b^2 B_d + q)I$ until the edge observation, which bounds the node at distance $d+1$ by $B_{d+1} I$; later observations only decrease it. This holds for every edge ordering, using the preceding period's bound. □

Full-coordinate pair sums have the same transfer bound (change the sign of one node); more generally $h_i x_i + h_j x_j$ with $h_j \neq 0$ gives $p_j^+ \le (h_i/h_j)^2 p_i + R/h_j^2$.

What this means for robotics. Collaborative localisation mostly uses *differences* (relative poses). An unanchored graph of differences is never observable, since adding a common offset to all nodes is invisible, so it cannot be a counterexample to the conjecture; with an anchor, Proposition 9.1 proves boundedness. The risk lies in *sum-type or mixed-sign* channels on an odd cycle with no direct anchor. A physical example is two-way time-of-flight ranging between devices whose unknown internal delays b_i are estimated as states: each range measurement between devices i and j contains $b_i + b_j$, so ranging around a triangle of devices gives exactly the pair-sum structure of Theorem 4.1 for the delay states, observable without an anchor. Shared calibration offsets entering two sensors with the same sign behave in the same way. Our divergence results additionally require growth ($a > 1$) in those states; we do not claim that a specific deployed system has this property.

10 Related work

This section says what is known about filters that approximate or neglect correlations, and what is specific to the IsoKF recurrence.

The IsoKF line. The marginal-ratio cross-covariance correction comes from the decentralised collaborative localisation (DCL) of Luft and co-authors [9, 10] and was developed into isolated Kalman filtering in the dissertation [5] and the journal article [6], which also gives the conjecture. Distributed multi-robot localisation with a decomposed centralised filter goes back at least to Roumeliotis and Bekey [12].

What Luft et al. (2018) establish. In the journal article [10], a relative measurement between robots i and j updates the pair exactly, leaves other robots unchanged, and corrects each cross term by $\Sigma_{ik}^{+} = \Sigma_{ii}^{+}(\Sigma_{ii})^{-1}\Sigma_{ik}$ [10, eq. (26)]; for scalar nodes this is exactly (2.3). Their measurement models are generic, so pair sums are admissible. They also introduce the scaled variant $\Sigma_{ik}^{+} = \lambda\,\Sigma_{ii}^{+}(\Sigma_{ii})^{-1}\Sigma_{ik}$, with $\lambda = 0$ keeping only the most recent partner's correlation [10, eq. (28)]. Their Proposition 1 proves that if every two-robot submatrix is positive definite, every such update (and the scaled update with $|\lambda| \le 1$) keeps every two-robot submatrix positive definite. Lemma 1 and Corollary 1 bound the one-step error of the cross-term approximation by $\|K_i F_j\|\sqrt{\epsilon_{ji}\epsilon_{ki}}$ in terms of conditional covariances. They note that any multiplicative cross-term update can underestimate cross-correlations, and they state that their estimator "is not provably consistent in general". On the UTIAS multi-robot dataset [10, Tables 2–4] the filter's accuracy is within about 12% of the centralised EKF, its normalised ANEES is below that of the EKF (it is more conservative), and its robustness is comparable to the EKF's. Luft et al. make no statement about boundedness, convergence or a steady state of the reported covariance over time, and give no example in which it diverges. Accordingly: the local well-posedness in Theorem 4.1 for pairwise schedules already follows from their Proposition 1; their scaled family with $0 \le \lambda \le 1$, including $\lambda = 0$, is a constant-factor case of Corollary 5.3; and their observation of conservatism on real data is consistent in direction with Finding 7.1, where the pessimism becomes unbounded. Since their correction is (2.3), Theorem 4.1 is equally an example of unbounded reported covariance for their filter with linear models; it contradicts no claim they make.

Citing works. A forward-citation check of [6] on 4 November 2026 found two related citing works, both read in full. The distributed Kalman-like observer of De Carli and Dimarogonas [2] replaces process noise by a forgetting factor and proves uniform global exponential convergence for that observer; it remarks that the filter of [10] lacks formal stability guarantees, but gives no divergence example. The buffer-driven estimator D-GVIO [11] uses IsoKF-style buffers, enforces consistency by covariance intersection and reports NEES experiments; it does not analyse IsoKF covariance convergence.

Consider (Schmidt) filters. Updating only some states while holding others fixed is the classical Schmidt–Kalman or consider filter [13]. Its exact cross-covariance correction for fixed nonparticipants is $F = CB^{-1} = I - KH$ applied to the whole participant block; IsoKF replaces it by the block-diagonal marginal ratio $\operatorname{diag}(C_{ii}B_{ii}^{-1})$, which avoids communicating the participants' joint factor. The full correction lies outside the class of Corollary 5.3; we prove no stability statement for it.

Correlation neglect and conservative fusion. Fusing estimates whose correlation is ignored can double-count information and become overconfident; covariance intersection [4] and split covariance intersection [8] restore consistency by conservative bounds. IsoKF sits between these: it keeps an approximate correlation rather than bounding it. Our failure is in the opposite direction from double counting: the reported covariance becomes far too pessimistic. Corollary 5.3 shows that the classical remedy of shrinking or deleting cross-covariances does not restore boundedness for the same models, and Finding 7.2 shows that deletion is in fact more robust than the IsoKF rule.

Graphical models and limits. The odd-cycle (non-bipartite) condition echoes known pathologies of loopy Gaussian belief propagation, whose means are correct at convergence but whose variances are not [15]. Communication-dependent tracking limits for networked estimation are studied in [7]. Static maximum-determinant completion [3] concerns one consistent partial matrix and does not by itself give an invariant for successive local substitutions.

11 Verification, software and open questions

This section lists how each result was checked, what the software observation does and does not show, and what remains open.

Checks. Exact rational programs check four complete epochs of Theorem 4.1 together with the identities (4.4)–(4.5), the schedule matrix and its characteristic polynomial $\lambda(\lambda^2 - 4\lambda - 1)$, the witnesses of Theorem 8.1 and Appendix A, and the anchored identity with non-commuting blocks. Certificates C1–C3 use outward-rounded interval arithmetic (mpmath 1.3.0). Every script performs explicit checks that also run under `python -O` and exits with a nonzero code on failure; all outputs were regenerated in normal and `-O` mode and agree byte for byte. All scripts, certificates and outputs are archived with this paper at doi:10.5281/zenodo.23941896. Figure 2 illustrates Theorem 4.1; its infinite-horizon conclusion follows from the proof, not from the plot.

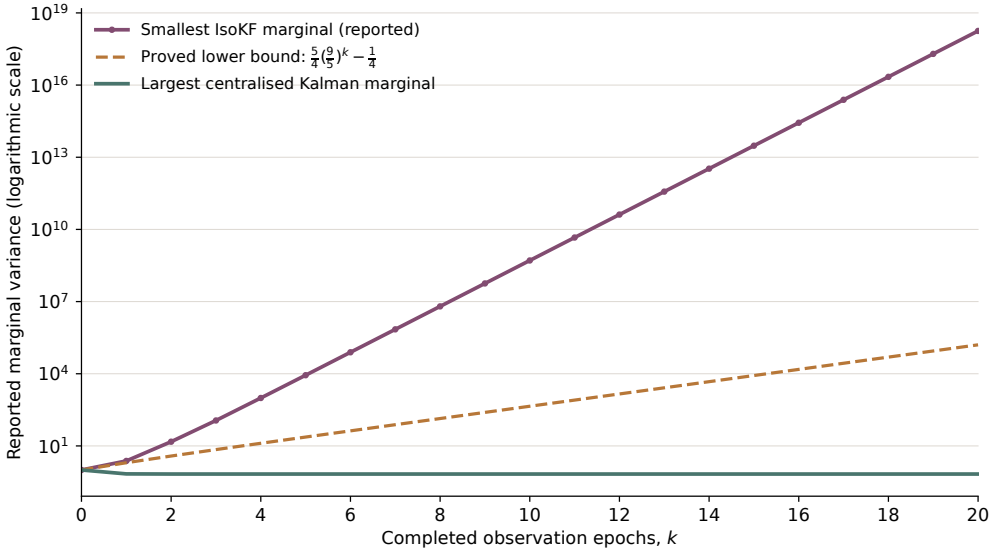


Figure 2: System of Theorem 4.1 ($A = 3I$, $Q = R = P_0 = I$), epochs 0–20, logarithmic scale. Solid purple: the *smallest* reported IsoKF marginal $\min_i P_{ii}$; dashed: the proved lower bound (4.2); green: the *largest* centralised Kalman marginal. Curves computed with 100-digit decimal arithmetic; the bound is analytic.

Software scope. The official `ikf_lib` source was inspected at a pinned commit [1]. Its observation handler includes numerical stabilisation, a check that the participant prior is positive semidefinite, and an optional projection of the participant posterior. Our pairwise counterexample keeps every participant prior and posterior positive definite, so the check would not trigger and the projection is inactive; it cannot mask the divergence. We did not compile or run the library and claim no observed software failure. The observation order is fixed and repeated, without delays; combining the three rows into one full-state update gives the ordinary Kalman filter and changes the architecture being tested.

Prior-art search. Two web engines, arXiv, the publisher's full text of [6], the full 2016 precursor [9], relevant chapters of [5], the software repository and a forward-citation sweep were searched on 1–4 November 2026. For version 0.2.0 the full 2018 journal article [10] and both citing works were read in full on 4 November 2026; Section 10 records what they establish. No matching theorem was found in the inspected corpus.

Changes in version 0.2.0. Version 0.1.0 compared this work with the 2018 journal article [10] at abstract level only. Version 0.2.0 compares it with the full text and credits that article's Proposition 1 (positive definiteness of two-robot blocks) and scaled correction (eq. (28)) where our results overlap; it also replaces title-level descriptions of the two citing works with full-text ones. No theorem, proof, certificate, computation or numerical finding has changed.

Open questions. (i) Entry into the escape region for every $a^2 \in (1, 1.38]$, and anything proved for $a^2 \in (1.38, 2+\sqrt{5})$. (ii) The exact divergence threshold from $P_0 = I$ (numerically $a^2 \approx 1.1225$). (iii) A proof that the actual estimation error stays bounded (Finding 7.1). (iv) Uniqueness of the steady state for Schur-stable A and the case $a = 1$. (v) Other graphs, noise ratios $Q \neq R$, and finite-precision behaviour of the software.

12 Conclusions

The convergence conjecture for isolated Kalman filtering is false. A three-node system with pairwise sum observations, observable from one epoch and controllable through its process noise, has reported variances that provably grow without bound while every local block stays positive definite and the centralised filter stays bounded. Divergence needs instability, but only a little: computer-assisted proofs show growth at the open-loop rate for large priors at instabilities down to $a^2 = 1 + 10^{-8}$, and coexistence of a bounded and a divergent trajectory at $a^2 = 21/20$, so there is no globally attracting steady covariance either. In the tested divergent cases the actual error of the IsoKF estimate stayed bounded (numerical finding). The practical failure is therefore that the reported covariance becomes grossly pessimistic and loses its meaning, not that the estimate diverges. Local positive definiteness is guaranteed exactly when participant groups overlap in at most one node, and anchored relative measurements remain safe.

A Exact failures outside the algorithm's domain

This appendix is outside the domain of the IsoKF update: it shows how unrestricted group processing can reach an array where the next update is undefined or meaningless. It is not needed for, and is not part of, the refutation.

Take $A = 2I_3$, $Q = I_3$, $P_0 = I_3$ and process $h_1 = (1, -1, 0)$, $h_2 = (0, 1, -1)$, $h_3 = (2, -3, 2)$ with noise variances $1, 1, 155/132$; the stacked matrix has determinant one. After prediction and the first two observations,

$$P_* = \begin{pmatrix} 30/11 & 25/16 & 0 \\ 25/16 & 15/8 & 25/16 \\ 0 & 25/16 & 205/96 \end{pmatrix}, \qquad \det P_* = -\frac{257375}{270336}.$$

Every two-node block is positive definite, but the prior for the three-participant observation has left the positive definite domain of Section 2; what follows is unchecked algebraic continuation of (2.2). Here $P_* h_3^\mathsf{T} = (135/176, 5/8, -5/12)^\mathsf{T}$ and $h_3 P_* h_3^\mathsf{T} = -155/132$, so the third innovation is exactly zero with nonzero numerator. With $R_3 = 5/4$ the innovation is $5/66 > 0$ but the marginals become $(-645/128, -105/32, -5/32)$. Writing $R_3 = 155/132 + \varepsilon$ gives $S = \varepsilon$ and new diagonal entries $(P_*)_{ii} - (P_* h_3^\mathsf{T})_i^2/\varepsilon$; all three are negative exactly when $0 < \varepsilon < 10/123$, i.e. $155/132 < R_3 < 2265/1804$. These strict inequalities persist under small perturbations within the fixed participant architecture.

B A positivity-preserving attenuation that still diverges

This appendix records a standard Schur-complement consequence. We claim no novelty for the matrix criterion and infer no stability or consistency from it.

Suppose a participant update replaces $B \succ 0$ by $C \succ 0$, leaves the nonempty nonparticipant block Z unchanged and replaces the cross block X by FX. The map $\left(\begin{smallmatrix} B & X \\ X^\mathsf{T} & Z \end{smallmatrix}\right) \mapsto \left(\begin{smallmatrix} C & FX \\ X^\mathsf{T}F^\mathsf{T} & Z \end{smallmatrix}\right)$ preserves positive semidefiniteness for every globally positive semidefinite completion (X, Z) if and only if

$$FBF^\mathsf{T} \preceq C. \tag{B.1}$$

Sufficiency: add $\operatorname{diag}(C - FBF^\mathsf{T}, 0)$ to a congruence of the original array. *Necessity:* one outside coordinate suffices. For a vector x take cross column x and outside variance $x^\mathsf{T}B^{-1}x$; this completion is positive semidefinite (not definite). Its image must be positive semidefinite, and the Schur-complement condition gives $x^\mathsf{T}F^\mathsf{T}C^{-1}Fx \le x^\mathsf{T}B^{-1}x$ for all x, i.e. $F^\mathsf{T}C^{-1}F \preceq B^{-1}$, equivalent to (B.1) via $\|C^{-1/2}FB^{1/2}\| \le 1$. If strictly positive definite test completions are wanted, use $Z + \epsilon I$; the images are positive definite for each $\epsilon > 0$, and letting $\epsilon \downarrow 0$ the limit is positive semidefinite because the positive semidefinite cone is closed.

For IsoKF put $D = \operatorname{diag}_{i \in U}(C_{ii}B_{ii}^{-1})$. The largest common factor $\alpha \in [0, 1]$ with $F = \alpha D$ satisfying (B.1) is

$$\alpha_* = \min\left\{1,\ \lambda_{\max}(C^{-1/2}DBD^\mathsf{T}C^{-1/2})^{-1/2}\right\}.$$

It uses only participant information; applying $\alpha_* D_i$ preserves global positive definiteness under any schedule from a globally positive definite start, and so prevents the failures of Appendix A. But by Corollary 5.3 it still diverges on the examples of Theorems 4.1–5.1: global positivity and boundedness are separate requirements. The full Schmidt correction $F = CB^{-1}$ also satisfies (B.1), since $CB^{-1}C \preceq C$, but lies outside Corollary 5.3.

References

[1] Control of Networked Systems, University of Klagenfurt. ikf_lib. GitHub repository, pinned commit 4c0d0ec4600b663cb116698d514e055fb371919d, 2025. URL https://github.com/aau-cns/ikf_lib/tree/4c0d0ec4600b663cb116698d514e055fb371919d. Source inspected 2 November 2026; not compiled or run.

[2] Nicola De Carli and Dimos V. Dimarogonas. A Distributed Kalman-Like Observer With Dynamic Inversion-Based Correction for Multi-Agent Estimation. *IEEE Control Systems Letters*, 9:523–528, 2025. doi: 10.1109/lcsys.2025.3576671.

[3] Dan I. Georgescu, Nicholas J. Higham, and Gareth W. Peters. Explicit solutions to correlation matrix completion problems, with an application to risk management and insurance. *Royal Society Open Science*, 5(3):172348, 2018. doi: 10.1098/rsos.172348.

[4] S.J. Julier and J.K. Uhlmann. A non-divergent estimation algorithm in the presence of unknown correlations. In *Proceedings of the 1997 American Control Conference (Cat. No.97CH36041)*, pages 2369–2373 vol.4. IEEE, 1997. doi: 10.1109/acc.1997.609105.

[5] Roland Jung. Recursive Distributed Collaborative Aided Inertial Navigation. arXiv:2408.12360, 2024. arXiv comment: PhD thesis.

[6] Roland Jung, Lukas Luft, and Stephan Weiss. Isolated Kalman filtering: theory and decoupled estimator design. *Autonomous Robots*, 49(1):7, 2025. doi: 10.1007/s10514-025-10191-x.

[7] Usman A. Khan and Ali Jadbabaie. Networked estimation under information constraints. arXiv:1111.4580, 2011. arXiv comment: Submitted to IEEE TAC.

[8] Hao Li and Fawzi Nashashibi. Cooperative multi-vehicle localization using split covariance intersection filter. In *2012 IEEE Intelligent Vehicles Symposium*, pages 211–216. IEEE, 2012. doi: 10.1109/ivs.2012.6232155.

[9] Lukas Luft, Tobias Schubert, Stergios I. Roumeliotis, and Wolfram Burgard. Recursive Decentralized Collaborative Localization for Sparsely Communicating Robots. In *Robotics: Science and Systems XII*. Robotics: Science and Systems Foundation, 2016. doi: 10.15607/rss.2016.xii.016.

[10] Lukas Luft, Tobias Schubert, Stergios I. Roumeliotis, and Wolfram Burgard. Recursive decentralized localization for multi-robot systems with asynchronous pairwise communication. *The International Journal of Robotics Research*, 37(10):1152–1167, 2018. doi: 10.1177/0278364918760698.

[11] Yarong Luo, Wentao Lu, Chi Guo, and Ming Li. D-GVIO: A Buffer-Driven and Efficient Decentralized GNSS-Visual-Inertial State Estimator for Multi-Agent Systems. arXiv:2603.01404, 2026. arXiv comment: Accepted by ICRA 2026.

[12] S.I. Roumeliotis and G.A. Bekey. Distributed multirobot localization. *IEEE Transactions on Robotics and Automation*, 18(5):781–795, 2002. doi: 10.1109/tra.2002.803461.

[13] Stanley F. Schmidt. Application of State-Space Methods to Navigation Problems. In *Advances in Control Systems*, pages 293–340. Elsevier, 1966. doi: 10.1016/b978-1-4831-6716-9.50011-4.

[14] Siew Chan, G. Goodwin, and Kwai Sin. Convergence properties of the Riccati difference equation in optimal filtering of nonstabilizable systems. *IEEE Transactions on Automatic Control*, 29(2):110–118, 1984. doi: 10.1109/tac.1984.1103465.

[15] Yair Weiss and William T. Freeman. Correctness of Belief Propagation in Gaussian Graphical Models of Arbitrary Topology. *Neural Computation*, 13(10):2173–2200, 2001. doi: 10.1162/089976601750541769.